Zachary Espiritu

New York, NY, USA

me@zacharyespiritu.com https://zacharyespiritu.com

Research Interests

Algorithms and data structures, cryptography, databases, formal methods and optimization, systems security, usable security.

Education

Sep 2017 – Dec 2021 **Brown University, Sc.M. (Concurrent Masters)** in Computer Science (GPA: 4.0 / 4.0)

Sep 2017 – Dec 2021 **Brown University, Sc.B. with Honors** in Computer Science (GPA: 4.0 / 4.0)

Advisor: Roberto Tamassia

Current Affiliations

Feb 2023 – current **MongoDB Research**, New York, NY

Senior Research Engineer

Conducting research on encrypted search and database security. Serves as primary contact for engineering teams for advising and reviewing cryptographic deployments within MongoDB.

May 2026 – current **New York University**, New York, NY

Visiting Researcher

Hosted by Joseph Bonneau.

Publications

⁂ *denotes papers where authors are listed alphabetically.*

1. **Z. Espiritu**. "Snapshots of TEE-Based Encrypted Databases Leak Plaintext Histograms via Dendrochronology". In *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security* (The Hague, Netherlands). 2026.
[CCS'26]

⋆ Also accepted as a refereed talk at BlackHat Europe 2026.



2. **Z. Espiritu**, D. Cash. "Plaintext Recovery Against Post-Filtering Access Control". In *Proceedings of the 35th USENIX Security Symposium* (Baltimore, MD, USA). 2026.
[USENIX SEC'26 / Acceptance Rate: 13.2%]



3. L. N. Rosenbloom, S. Kamara, **Z. Espiritu**, T. Moataz, A. Bahi, J. Wilkinson. "tigro: Trust Infrastructure for Grassroots Organizing via Grounded Digital Annotations". In *Proceedings of Privacy Enhancing Technologies*, Vol. 2026, Iss. 2 (Calgary, Canada). 2026.
[PETS'26 / Acceptance Rate: 20%]

4. **Z. Espiritu** ⁂, S. Kamara, T. Moataz, V. Ogier. "Leafblower: a Leakage Attack Against TEE-Based Encrypted Databases". In *Proceedings of the 47th IEEE Symposium on Security and Privacy* (San Francisco, CA, USA). 2026.
[S&P'26 / Acceptance Rate: 12.7%]



5. M. George, S. Kamara, T. Moataz, **Z. Espiritu**. "Structured Encryption and Distribution-aware Leakage Suppression". In *31st International Conference on the Theory and Application of Cryptology and Information Security* (Melbourne, Australia). 2025.
[ASIACRYPT'25 / Acceptance Rate: 26.6%]

6. **Z. Espiritu** ⁂, S. Kamara, T. Moataz, A. Park. "PolySys: an Algebraic Leakage Attack Engine". In *Proceedings of the 34th USENIX Security Symposium* (Seattle, WA, USA). 2025.
[USENIX SEC'25 / Acceptance Rate: 17.1%]



7. A. Agarwal, **Z. Espiritu** ⁂. "Sequentially Consistent Concurrent Encrypted Multimaps". In *Proceedings of the 10th IEEE European Symposium on Security and Privacy* (Venice, Italy). 2025.
[EuroS&P'25 / Acceptance Rate: 20%]

8. **Z. Espiritu** ⁂, S. Kamara, T. Moataz. "Bayesian Leakage Analysis: A Framework for Analyzing Leakage in Cryptography". In *IACR Communications in Cryptology*, Vol. 2, Iss. 1. 2025.
[CIC'25 / Acceptance Rate: 32%]

9. **Z. Espiritu** ⁂, M. George, S. Kamara, L. Qin. "Synq: Public Policy Analytics Over Encrypted Data". In *Proceedings of the 45th IEEE Symposium on Security and Privacy* (San Francisco, California, USA). 2024.
[S&P'24 / Acceptance Rate: 17.8%]

10. F. Falzon, E. A. Markatou, **Z. Espiritu**, R. Tamassia. "Range Search over Encrypted Multi-Attribute Data". In *Proceedings of the VLDB Endowment*, Vol. 16, Iss. 4. (Vancouver, Canada). 2023.
[VLDB'23 / Acceptance Rate: 24.8%]



11. E. A. Markatou, F. Falzon, **Z. Espiritu**, R. Tamassia. "Attacks on Encrypted Response-Hiding Range Search Schemes in Multiple Dimensions". In *Proceedings of Privacy Enhancing Technologies*, Vol. 2023, Iss. 2. (Lausanne, Switzerland). 2023.
[PETS'23 / Acceptance Rate: 22%]

12. **Z. Espiritu**, E. A. Markatou, R. Tamassia. "Time- and Space-Efficient Aggregate Range Queries on Encrypted Databases". In *Proceedings of Privacy Enhancing Technologies*, Vol. 2022, Iss. 4. (Sydney, Australia). 2022.
[PETS'22 / Acceptance Rate: 24%]

Patents / Applications

13. A. Agarwal, **Z. Espiritu** ⁂, M. George, S. Kamara, T. Moataz, A. Park, inventors; MongoDB, Inc., assignee. "Systems and Methods for Multi-Party Private Database Implementation". 2025. *Patent pending.* `[US-20250358112-A1]`

Previous Employment

Feb 2022 – Feb 2023 — **D.E. Shaw & Co.**, New York, NY
Systems Security Engineer

Summer 2021 — **Cloud Security Group at Brown University**, Providence, RI
Undergraduate Researcher

Summer 2020 — **Google**, Remote
Software Engineering Intern, EKM (External Key Manager) / KMS (Key Management Service)

Summer 2018 — **Brown PLT at Brown University**, Providence, RI
Undergraduate Researcher

Summer 2017 and Summer 2016 — **Order.co (formerly Negotiatus)**, New York, NY
Software Engineering Intern

Awards and Funding

See "Professional Service" below for reviewing-related awards.

2022 *Brown CS Norman K. Meyrowitz '81 Award*

Awarded for "exceptionally meritorious service to Brown CS". (2nd student to receive in award history; one of 4 students to receive this award since it was created in 2020.)

Brown CS Senior Prize

Awarded for academic work and service to Brown CS (6.8% of graduating CS undergraduates).

2021 *Crowdstrike NextGen Scholarship* ($10,000)

Brown CS Randy Pausch Undergraduate Research Fellowship ($10,000)

(ISC)² Undergraduate Cybersecurity Scholarship ($3,000)

Teaching Assistant Experience

Courses were taught at Brown University.

Fall 2021	CSCI 1650: Software Security and Exploitation (Head TA)
Spring 2021	CSCI 1660: Computer Systems Security (Head TA)
Fall 2020	CSCI 1730: Design and Implementation of Programming Languages (Head TA)
Spring 2020	CSCI 1660: Computer Systems Security (Head TA)
Fall 2019	CSCI 1730: Design and Implementation of Programming Languages
Spring 2019	CSCI 1660: Computer Systems Security (Head TA)
Fall 2018	CSCI 0190: Accelerated Introduction to Computer Science (Head TA)
Spring 2018	CSCI 0040: Introduction to Scientific Computing and Problem-Solving
Fall 2017	CSCI 0050: A Data-Centric Introduction to Programming

Research Mentoring / Supervision

2026 Undergraduate intern Yoav Rafalin at MongoDB Research.

2024 Masters intern Valentin Ogier at MongoDB Research, *now at NetApp*. Collaboration led to *Leafblower* [4] at IEEE S&P 2026.

Undergraduate intern John Wilkinson at MongoDB Research, *now at Saronic*. Collaboration led to *Tigro* [3] at PETS 2026.

Professional Service

Program Committee USENIX Security: 2027
PETS: 2027

Artifact Evaluation ACM CCS: 2026
IEEE S&P: 2026
USENIX Security: 2026 (**Distinguished Reviewer Award**, **Ninja Reviewer Recognition**)

PETS: 2026 (**Distinguished Artifact Reviewer**), 2027

External Reviewer

ACM CCS: 2023
CRYPTO: 2023, 2024, 2025
PETS: 2027
SRDS: 2024
VLDB: 2025

Journal Reviewer

ACM TOPS: 2024

Invited Talks

Dec 2026 — New York University, *CryptoSec Seminar*, "Snapshots of TEE-Based Encrypted Databases Leak Plaintext Histograms via Dendrochronology" *(based on [1])*

Oct 2026 — University of Chicago, *Topics in Computer Science Seminar*, "Snapshots of TEE-Based Encrypted Databases Leak Plaintext Histograms via Dendrochronology" *(based on [1])*

May 2026 — Brown University, *Guest Lecture in CSCI1710: Logic for Systems*, "Messy, Real-World Cryptography (and How Formal Methods Can Help)" *(based on [6] and [8])*

Apr 2026 — Brown University, *Theory Seminar*, "Leafblower: a Leakage Attack Against TEE-Based Encrypted Databases" *(based on [4])*
`https://www.youtube.com/watch?v=UE471UInufc`

Jan 2026 — University of Chicago, *Topics in Computer Science Seminar*, "Leafblower: a Leakage Attack Against TEE-Based Encrypted Databases" *(based on [4])*

Organization Service

MongoDB Research, New York, NY

Aug 2024 – current — *MongoDB PhD Fellowship Committee*

Department of Computer Science, Brown University, Providence, RI

Oct 2019 – Dec 2021 — *Meta Teaching Assistant*

Coordinated hiring and training of 600+ teaching assistants each year across 50 CS courses in collaboration with department faculty and staff.

Apr 2019 – Dec 2021 — *SPOC (Systems Programmer, Operator, Consultant) / Sunlab Consultant*

On-call technical staff supporting all Linux infrastructure in the department.

Interests and Hobbies

Marathon (and ultramarathon) running

Pottery on the wheel

Theatrical lighting design

Security CTF competitions (solo winner of Brown's 2021 CTF and MongoDB's 2024 CTF)
Dance Dance Revolution